A. El fucho

time limit per test: 1 second

memory limit per test: 256 megabytes

Juan and his friends are going to split themselves into n teams and play a modified double-elimination football tournament, consisting of a *winners'* group and a *losers'* group. Initially, all teams belong to the winners' group.

In each round of the tournament, the following happens as long as one of the groups has **at least two** teams:

- All teams in the winners' group pair up.
  - If there is an odd number of teams in the winners' group, there would be a team that didn't get paired up (and wouldn't play a match). That team stays in the winners' group.
  - For teams in the winners' that got paired up, each pair plays a football match in which there are **no ties**.
    - If a team wins, it stays in the winners' group.
    - If a team loses, it drops down to the losers' group in the **next round**.
- All teams in the losers' group pair up.
  - If there is an odd number of teams in the losers' group, there would be a team that didn't get paired up (and wouldn't play a match). That team stays in the losers' group.
  - For teams in the losers' that got paired up, each pair plays a football match in which there are **no ties**.
    - If a team wins, it stays in the losers' group.
    - If a team loses, it gets eliminated from the tournament.

Note that in the above process, when a team loses a match in the winners' group, it drops down to the losers' group in the next round. That means, it is not considered for the pairing process in the current round's losers' group.

After multiple iterations of the aforementioned process, both groups end up with a single team each. When this happens, both teams face off against each other in a match and a victor emerges.

Determine how many matches were played in total. It can be proved that no matter how the teams are paired up and which ones win and lose, the answer remains the same.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 100$). The description of the test cases follows.

The only line of each test case contains one positive integer n ($2 \le n \le 500$) — the number of teams.

Output

For each test case, print the total number of matches played during the tournament.

Example

input Copy

```
2
2
```

Codeforces Round 1056 (Div. 2)

Contest is running

01:42:11

Contestant

→ Submit?

Language: Python 3.13.2

Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

Submit

→ Score table

	Score
Problem A	466
Problem B	932
Problem C	1631
Problem D	1864
Problem E	2097
Problem F	2796
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 00:17 from the first attempt

```
3
```

output

Copy

```
2
4
```

Note

In the first test case, it can be proved that exactly 2 matches are played before a victor is declared.

In the second test case, the image below shows one possible tournament. Notice that 4 matches were played in total.